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BAX 423

JobPilot Technical Brief

Overview

JobPilot is a job recommendation system designed to match job seekers with relevant employment opportunities using semantic retrieval, user-defined filtering, multi-factor ranking, adaptive feedback learning, and generative AI. The system was developed as a final project for BAX 423 and demonstrates recommendation-system techniques including embeddings, approximate nearest-neighbor retrieval, re-ranking, explainability, adaptive learning, and resume personalization. The recommendation workflow consists of six major stages: job ingestion and deduplication, profile intake and skill extraction, embedding-based retrieval, multi-stage ranking, adaptive feedback learning, and recommendation export.

Capability 1: Job Data Ingestion and Deduplication

The original source dataset consisted of a 46.9 GB newline-delimited JSON file. For reproducible submission, a smaller raw JSONL sample was created and included in the project folder. The notebook streams this sample line by line, standardizes records, deduplicates postings, and creates the processed job snapshot used downstream. The resulting dataset serves as the foundation for retrieval, ranking, analytics, and recommendation generation.

Capability 1.1: Live Data

JobPilot currently uses the large preprocessed 25,000 job-posting dataset as the primary retrieval source. To support future real-time job search, an optional Adzuna API ingestion module was added to the notebook as a demonstration. This module retrieves current job postings, standardizes fields into the JobPilot schema, deduplicates records, and saves a live-job snapshot. The deployed Streamlit app uses the stable 25,000-job snapshot and FAISS index for reproducible recommendation results.

Capability 2: Profile Intake and Skill Matching

JobPilot converts uploaded resumes into structured candidate profiles. Resume text is extracted from PDF documents and matched against a controlled skill dictionary containing technical, analytical, and business-related skills. The extracted profile includes candidate skills, target roles, location preferences, and dealbreakers. These structured features support downstream filtering, ranking, recommendation explanations, and resume generation.

Capability 3: Embedding-Based Job Retrieval

Both job postings and candidate profiles are represented using dense sentence embeddings generated with SentenceTransformers. Job embeddings are normalized and indexed using FAISS, enabling efficient approximate nearest-neighbor retrieval across the 25,000-posting dataset. Rather than ranking every posting directly, JobPilot first retrieves the most semantically relevant candidates and passes them to downstream filtering and ranking stages. Embedding generation is performed offline and stored as precomputed artifacts, allowing the deployed application to load embeddings and retrieval indexes without recomputation. Manual evaluation of the top retrieved candidates indicated approximately 80% relevance within the top ten recommendations.

Capability 4: Multi-Stage Ranking Pipeline

After candidate generation, JobPilot applies user-defined hard filters and a multi-factor ranking model. Hard filters remove jobs that violate user-defined constraints such as seniority mismatch, internship-only positions, unpaid roles, or contract-only opportunities. Filtering logic adapts to user experience level to avoid applying identical rules across all candidates.

Remaining jobs are scored using a weighted ranking model that combines:

- Semantic similarity
- Skill overlap
- Target-role alignment
- Location alignment

The re-ranking stage improves recommendation quality on top of pure embedding similarity by incorporating user-specific preferences and interpretable matching signals.

Capability 5: Adaptive Learning from Feedback

JobPilot implements adaptive preference learning using Accept, Reject, and Skip feedback. Feedback is used to update ranking weights associated with semantic similarity, skill overlap, role alignment, and location alignment. After each update, recommendations are re-ranked to reflect observed user preferences. Adaptive learning was evaluated using simulated interaction rounds. After running a simulation, the average accepted-job rank improved from 5.75 to 5.00 over two feedback iterations, representing approximately 13% improvement in recommendation quality. The system learned to increase the importance of role alignment while reducing reliance on less informative features. The deployed Streamlit application also supports interactive feedback collection and live recommendation re-ranking during a user session.

Capability 6: Recommendation Export

Users can export their top recommendations as a downloadable CSV file containing job title, employer, location, recommendation score, explanation, and application link. This functionality enables users to save and review recommended opportunities outside the application.

Generative Resume Tailoring

In addition to recommendation functionality, JobPilot integrates Google Gemini 2.5 Flash to generate tailored resume drafts for selected job postings. The model receives the candidate's original resume and the selected job description, then produces a revised resume emphasizing relevant skills and keywords while preserving factual experience. This feature improves ATS alignment and provides users with a customized application document.

Analytics Dashboard

JobPilot includes a batch analytics layer that summarizes demand patterns across the 25,000-posting dataset.

The dashboard reports:

- Frequently requested skills
- High-volume hiring locations
- Major employers
- Common job titles
- Salary-data coverage

Because salary information was unavailable in the source dataset, salary analysis is reported as a dataset limitation.

Test Persona Results

Persona 1: New Graduate

As a soon-to-be MSBA graduate with skills in Python, SQL, Power BI, and Databricks, I uploaded my resume and JobPilot successfully retrieved and ranked entry-level Data Analyst, Business Analyst, and Analytics Engineer positions that matched my technical background. As my resume included experience in manufacturing, process improvement, Python, SQL, Databricks, and analytics coursework, JobPilot successfully retrieved and ranked relevant analyst and analytics engineering positions. The system demonstrated strong semantic matching, identified transferable technical skills, and improved recommendation quality through adaptive feedback updates. Additional evaluation considered career pivoter, experienced specialist, and location-constrained scenarios, demonstrating that the same recommendation architecture can support multiple job-seeking profiles.

Persona 2: Career Pivoter

A candidate transitioning from Engineering into Data Analytics will upload a resume emphasizing process improvement, reporting, and technical problem-solving. The system will identify analyst positions where transferable skills complemented analytics requirements.

Persona 3: Experienced / Niche Candidate

A candidate with specialized experience in data engineering and cloud platforms will submit a resume targeting Analytics Engineer roles. JobPilot can prioritize positions requiring Databricks, SQL, and large-scale data processing skills.

Persona 4: International / Visa-Constrained Candidate

A candidate prioritizing remote opportunities and location flexibility can use JobPilot to identify roles with broader geographic availability. The location filtering stage can successfully reduce recommendations that did not meet stated preferences.

Limitations

Several limitations are:

- The dataset represents a September 2021 snapshot and may not reflect current job availability.
- Salary information was largely unavailable in the source data.
- Resume-generation quality depends on Gemini output quality.
- Adaptive learning is session-based and therefore does not carry over to future users or future sessions.

Conclusion

The completed JobPilot system successfully demonstrates an end-to-end recommendation workflow that combines information retrieval, machine learning, adaptive ranking, explainability, and generative AI. The deployed Streamlit application supports resume intake, semantic retrieval, multi-stage ranking, adaptive feedback learning, recommendation export, analytics reporting, and AI-assisted resume generation within a unified user-facing platform.